

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397963
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Dunn; Dave et al.

Container assembly having a heat-sealed metal end, a metal end therefor, and a method for making same

Abstract

A container assembly includes a container body having a side wall encircling an axis, and a metal end for attachment to an upper edge of the side wall via heat-sealing. The outer peripheral region is shaped prior to application to the container body such that an annular channel is defined between an inner chuck wall and an outer chuck wall of the metal end. The metal end is pushed straight onto the side wall such that the upper edge of the side wall is received into the channel. The surfaces of the side wall and the opposing surfaces of the chuck walls have heat-sealable material thereon. The metal end is heated to melt and fuse the heat-sealable layers, thereby sealing the metal end onto the side wall. The metal end is shaped such that the free edge of the outer chuck wall is not exposed.

Inventors: Dunn; Dave (Carrollton, OH), Meeks; John (Massillon, OH), Morin; Jeremy (Zionsville, IN), Price; Trevor (North Canton, OH)

Applicant: SONOCO DEVELOPMENT, INC. (Hartsville, SC)

Family ID: 1000008781328

Assignee: SONOCO DEVELOPMENT, INC. (Hartsville, SC)

Appl. No.: 18/123030

Filed: March 17, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230219714 A1	Jul. 13, 2023

Related U.S. Application Data

continuation parent-doc US 14846250 20150904 US 9789996 20171017 child-doc US 15707645
continuation parent-doc US 12607731 20091028 US 9150328 20151006 child-doc US 14846250

Publication Classification

Int. Cl.: **B65D8/20** (20060101); **B65D6/28** (20060101); **B65D6/30** (20060101); **B65D8/00** (20060101); **B65D21/02** (20060101)

U.S. Cl.:

CPC **B65D7/36** (20130101); **B65D7/34** (20130101); **B65D15/06** (20130101); **B65D15/08** (20130101); **B65D21/0209** (20130101); **B65D21/0222** (20130101); B65D2251/20 (20130101); Y10T29/49826 (20150115)

Field of Classification Search

CPC: B65D (15/18); B65D (15/08); B65D (15/06); B65D (7/36); B65D (21/0209); B65D (21/0222); B65D (21/022)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1552134	12/1924	Frey	N/A	N/A
1876450	12/1931	Egan et al.	N/A	N/A
2176950	12/1938	Chester	N/A	N/A
2237809	12/1940	Bronson	N/A	N/A
2244341	12/1940	Maclean	N/A	N/A
2445647	12/1947	Thomas	N/A	N/A
2899096	12/1958	Henchert et al.	N/A	N/A
2901161	12/1958	Henchert	N/A	N/A
3108710	12/1962	Lange et al.	N/A	N/A
3122274	12/1963	Quinche	N/A	N/A
3173574	12/1964	Goldsmith	N/A	N/A
3206060	12/1964	Turner	N/A	N/A
3228552	12/1965	Rutledge	N/A	N/A
3286867	12/1965	Mumford	N/A	N/A
3288342	12/1965	Tinker	N/A	N/A
3306488	12/1966	Lemelson	N/A	N/A
3315865	12/1966	Bracht	N/A	N/A
3367533	12/1967	Baker	N/A	N/A
3378163	12/1967	Griffith	N/A	N/A
3406891	12/1967	Buchner et al.	N/A	N/A
3424343	12/1968	Hoeffelman	N/A	N/A
3580464	12/1970	Griffith	N/A	N/A
3615033	12/1970	Podesta	N/A	N/A
3620437	12/1970	Kulp	N/A	N/A
3624789	12/1970	Peyser et al.	N/A	N/A
3721365	12/1972	Saunders	N/A	N/A

3773208	12/1972	Curry	N/A	N/A
3958717	12/1975	Ellis	N/A	N/A
4044941	12/1976	Knudsen	N/A	N/A
4094460	12/1977	Scanga et al.	N/A	N/A
4252265	12/1980	Brundige et al.	N/A	N/A
4254886	12/1980	Van Doren	N/A	N/A
4356926	12/1981	Priestly et al.	N/A	N/A
4363404	12/1981	Westphal	N/A	N/A
4372719	12/1982	Supik	N/A	N/A
4376506	12/1982	Westphal	N/A	N/A
4448345	12/1983	Helms	N/A	N/A
4538758	12/1984	Griffith	N/A	N/A
4597502	12/1985	Troughton	N/A	N/A
4650088	12/1986	Hirota et al.	N/A	N/A
4655359	12/1986	Fairgrieve	N/A	N/A
4692132	12/1986	Ikushima et al.	N/A	N/A
4754113	12/1987	Mohr et al.	N/A	N/A
4757175	12/1987	Mohr et al.	N/A	N/A
4785992	12/1987	Goepfner	N/A	N/A
4853510	12/1988	Mohr et al.	N/A	N/A
4948006	12/1989	Okabe et al.	N/A	N/A
5152418	12/1991	Kroeschell et al.	N/A	N/A
5169017	12/1991	Cooper et al.	N/A	N/A
5199593	12/1992	Kita	N/A	N/A
5427268	12/1994	Downing, Jr. et al.	N/A	N/A
5433992	12/1994	Galda et al.	N/A	N/A
5971259	12/1998	Bacon	N/A	N/A
5993593	12/1998	Swartz et al.	N/A	N/A
5996835	12/1998	Farrington et al.	N/A	N/A
6079185	12/1999	Palaniappan et al.	N/A	N/A
6082541	12/1999	Bewick	N/A	N/A
6126029	12/1999	Storgaard	N/A	N/A
6194041	12/2000	McHenry et al.	N/A	N/A
6253951	12/2000	Pruckler	N/A	N/A
6258312	12/2000	Heyn	N/A	N/A
6362461	12/2001	Wiening et al.	N/A	N/A
6427862	12/2001	Hsu	N/A	N/A
6774560	12/2003	Smith et al.	N/A	N/A
6883689	12/2004	Odet	N/A	N/A
7380685	12/2007	Simmons et al.	N/A	N/A
8998027	12/2014	Price et al.	N/A	N/A
9150328	12/2014	Dunn et al.	N/A	N/A
9789996	12/2016	Dunn et al.	N/A	N/A
10532851	12/2019	Dunn et al.	N/A	N/A
11628969	12/2022	Dunn	220/619	B65D 7/36
2002/0125249	12/2001	Baird-Smith et al.	N/A	N/A
2003/0168423	12/2002	Williams	N/A	N/A
2004/0069790	12/2003	Alexander et al.	N/A	N/A
2007/0284276	12/2006	Luttik et al.	N/A	N/A
2008/0050206	12/2007	Kwakkel et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
669 574	12/1988	CH	N/A
0 036 763	12/1980	EP	N/A
0 264 946	12/1987	EP	N/A
0 276 472	12/1987	EP	N/A
0 332 725	12/1988	EP	N/A
0 536 137	12/1992	EP	N/A
0 537 753	12/1992	EP	N/A
0 868 371	12/1997	EP	N/A
1 889 673	12/2007	EP	N/A
WO 00/06369	12/1999	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion for International Application No. PCT/US2010/052412, mailed Dec. 28, 2010. cited by applicant

Primary Examiner: Castellano; Stephen J

Attorney, Agent or Firm: Nelson Mullins Riley & Scarborough LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 16/703,411, filed Dec. 4, 2019, which is a divisional of U.S. application Ser. No. 15/707,645, filed Sep. 18, 2017, which is a continuation of U.S. Ser. No. 14/846,250, filed Sep. 4, 2015, which is a continuation of U.S. application Ser. No. 12/607,731, filed on Oct. 28, 2009, both of which are hereby incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

(1) The present disclosure relates to containers in general, and more particularly to containers that are closed by a metal end seamed onto the container body.

(2) For many years the standard technique for attaching a metal end to a metal container body has been the double-seaming method, in which a curled outer edge of the metal end and a flange formed on the container body are rolled up together to form a hermetic seam. The double-seaming technique essentially entails mechanically locking together the metal end and the container body flange. Double-seaming works well for metal container bodies because the metal is ductile and permanently deformable without compromising the integrity of the seam, but encounters difficulties when the container body is plastic or composite because of the relative lack of ductility and lack of permanent deformability of such materials.

BRIEF SUMMARY OF THE DISCLOSURE

(3) The present disclosure relates to an alternative to the traditional double-seaming approach for attaching metal ends to containers. The metal end and the method described herein for affixing the metal end to a container body are suitable for container bodies of various materials including metal, plastic, and composite paperboard/plastic or paperboard/foil/plastic materials.

(4) In accordance with one embodiment, a container assembly comprises a container body formed by a side wall having an upper edge, and a top metal end for affixing to the upper edge of the side wall. The metal end is formed of a shaped metal sheet that has a peripheral edge. The sheet is

shaped prior to being applied to the container body such that the outer peripheral region of the sheet includes an inner chuck wall that extends upwardly to a top rim of the metal end, the top rim comprising an outwardly curled, downwardly facing upper U-shaped section. The outer peripheral region of the metal sheet is further shaped to include a skirt extending downwardly from the top rim, a lower end portion of the skirt being formed as an inwardly curled, upwardly facing lower U-shaped section, and an outer chuck wall extending upwardly from the lower U-shaped section, the peripheral edge of the metal sheet defining a top edge of the outer chuck wall. The outer chuck wall is radially spaced from the inner chuck wall so as to define an annular channel therebetween configured to allow the top metal end to be pushed onto the upper edge of the side wall such that the upper edge is received into the annular channel. A heat-sealable material is disposed on at least one of the opposing surfaces of the inner and outer chuck walls.

(5) The metal end is sealed to the container body not by a purely mechanical locking together of the metal end and container body as in conventional double-seaming, but instead by thermal fusion of the heat-sealable material provided on the inner and/or outer surfaces of the side wall and on the opposing surfaces of the inner and/or outer chuck walls. Indeed, the step of pushing the metal end onto the container body to insert the upper edge of the side wall into the annular channel does not require or involve any significant deformation of the metal end or of the container body side wall. Consequently, the material of the container body and its mechanical properties are not of particular importance to the proper attachment of the metal end, and hence the approach is suitable for container bodies of virtually any material as long as a heat-sealable material can bond to the container body with sufficient strength.

(6) Metal ends can be affixed in this manner to both top and bottom ends of a generally tubular container body (for example, a composite paperboard/plastic or paperboard/foil/plastic can), if desired. The approach is also useful for container bodies such as blow-molded plastic cans that require only a single metal end for closing the top opening.

(7) The metal ends can be formed to allow containers to be stacked.

(8) The heat-sealing of the metal end onto the container body can be accomplished in various ways, including but not limited to induction heating of the metal end to melt and fuse the heat-sealable materials together.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

(1) Having thus described the disclosure in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

(2) FIG. 1 is a fragmentary cross-sectional view of a container assembly in accordance with one embodiment of the invention, showing a metal end prior to being applied onto the upper edge of the container body side wall;

(3) FIG. 2 is a view similar to FIG. 1, showing the metal end after it has been pushed onto the side wall of the container body;

(4) FIG. 3 is a view similar to FIG. 2, showing a bottom of a second container stacked atop the metal end of the first container; and

(5) FIG. 4 is a view similar to FIG. 2, showing another embodiment of the invention.

DETAILED DESCRIPTION OF THE DRAWINGS

(6) The present invention now will be described more fully hereinafter with reference to the accompanying drawings in which some but not all embodiments of the inventions are shown. Indeed, these inventions may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like numbers refer to like elements

throughout.

(7) A container assembly **10** in accordance with a first embodiment of the invention is shown in FIG. **1**. The container assembly comprises a container body **12** and a metal end **30**. The container body **12** can comprise any of various configurations and materials. Generally, the container body will include a side wall **14** that encircles an axis. The side wall **14** has an upper edge **16** that circumscribes a top opening **18** of the container body. The side wall can define a cross-sectional shape for the container body that is round, oval, oblong, rectangular, square, triangular, or any other desired shape, and thus the top opening **18** can have any of such shapes. In the specification and claims of the present application, references to the “radial” direction are not meant to suggest or require that the container body has a round cross-sectional shape, but rather refer more generally to the direction perpendicular to the axis about which the side wall **14** extends, and thus apply to any cross-sectional shape for the container body.

(8) The side wall **14** can be formed of various materials, including metal, plastic, or composite materials. By “composite” material is meant a multi-ply structure formed as a laminate of two or more layers of different materials, typically including at least one paperboard layer and at least one additional layer that can comprise metal foil such as aluminum foil or a plastic layer such as plastic film or a polymer coating. Often, composite containers include paperboard, foil, and plastic layers. For example, a typical composite container as illustrated in FIG. **1** has a side wall **14** of paperboard plies **20** adhesively laminated together to provide structural strength to the container body, and an impervious liner **22** adhered to the radially inner surface of the wall formed by the paperboard plies for rendering the side wall **14** substantially impervious to liquids and gases. The liner **22** typically includes a barrier layer such as metal foil or polymer film (e.g., metallized polyester, EVOH, SiO.sub.x-coated polyester, AlO.sub.x-coated polyester, or the like). The particular structure of the liner is not critical to the present invention. The composite side wall typically also includes an outer label ply **24** adhered to the radially outer surface of the wall formed by the paperboard plies. The label ply can comprise a thin paper layer (e.g., thin kraft), a plastic film layer, or the like.

(9) The radially innermost surface of the side wall **14** has a layer **26** of heat-sealable material covering at least the part of the side wall adjacent the upper edge **16**. In the embodiment shown in FIG. **1**, the heat-sealable layer **26** covers the entire inner surface of the side wall, but alternatively the heat-sealable layer could be localized to the region adjacent the upper edge **16**. The heat-sealable layer **26** can comprise a layer of the liner **22** that is employed for sealing edges of the liner together to form a joint or seam for the liner. Alternatively, the heat-sealable layer **26** can comprise a coating provided specifically for purposes of attaching the metal end **30** to the side wall as described below.

(10) The radially outermost surface of the side wall **14** likewise has a layer **28** of heat-sealable material covering at least the part of the side wall adjacent the upper edge **16**. The heat-sealable layer **28** can be localized to the region adjacent the upper edge **16** or can cover the entire outer surface of the side wall (e.g., the layer **28** could be an outer layer of the label ply **24** provided for sealing edges of the label ply together to form a joint or seam for the label). A localized heat-sealable layer **28** is shown in FIG. **1**.

(11) When the side wall **14** is not a composite material as described above, the heat-sealable materials **26**, **28** nevertheless are provided on the inner and outer surfaces at least adjacent the upper edge **16**. For example, if the side wall is metal, then a coating of heat-sealable material can be provided on each of the inner and outer surfaces. The coating on the inner surface can serve to protect the container contents from direct contact with the metal. If the side wall is plastic (e.g., when the container body is a blow-molded plastic container), the side wall material itself can comprise a heat-sealable material such as polypropylene or polyethylene.

(12) The metal end **30** is formed of a blank of metal sheet **32**. The metal sheet can have a thickness ranging from about 0.0055” to about 0.0110”. The sheet is stamped or cut out to have a plan shape generally corresponding to the cross-sectional shape of the container body side wall at the upper

edge **16**. Thus, if the upper edge **16** has a circular shape then the blank is circular, if the upper edge is rectangular then the blank is rectangular, etc. The metal sheet **32** has an outer peripheral edge **34**. The region of the metal sheet adjacent the outer edge **34** is deformed by suitable tool and die operations to have a configuration enabling the metal end **30** to be affixed to the side wall **14** in a manner described below.

(13) More particularly, the outer peripheral region of the metal sheet **32** is shaped to be annular (i.e., to encircle the central axis about which the side wall **14** extends, and to have a substantially uniform cross-sectional shape about the annulus). The annular shape of this peripheral region substantially matches that of the side wall **14** of the container body at the upper edge **16**. The peripheral region of the metal end **30** is shaped to include an inner chuck wall **36** that extends generally upwardly from the main central panel **38** of the metal end. The inner chuck wall can include a substantially linear upper portion **40** that is relatively closer to vertical (i.e., closer to being parallel to the central axis) but that can be somewhat inclined relative to vertical as shown in FIG. **1**, and a substantially linear lower portion **42** that is connected to the lower end of the upper portion **40** and that is relatively farther from vertical. The lower radially inner end of the lower portion **42** joins with the central panel **38**. The upper end of the upper portion **40** of the inner chuck wall joins with a top rim **44** of the metal end. The top rim **44** is shaped as a generally U-shaped section that faces downwardly (i.e., the open end of the “U” faces generally downwardly).

(14) The outer peripheral region of the metal sheet **32** is further shaped to include a skirt **46** that depends from the top rim **44** and that extends downwardly approximately parallel to the upper portion **40** of the inner chuck wall **36** and is spaced radially outwardly therefrom. The skirt in the embodiment of FIG. **1** includes a first (uppermost) portion **48** located just below the top rim **44**, a second (middle) portion **50** located below the first portion **48**, and a third (lowermost) portion **52** located below the second portion **50**. The second portion **50** is oriented closer to horizontal than the first and third portions **48**, **52** and forms a stacking surface on which a bottom of another container can be stacked, as further described below in connection with FIG. **3**.

(15) The lowermost or third portion **52** of the skirt joins with a lower generally U-shaped section **54** that faces generally upwardly. An outer chuck wall **56** extends upwardly from the lower U-shaped section **54** and is spaced from the inner chuck wall **36** (and specifically the upper portion **40** thereof). The lower end of the outer chuck wall **56** is at approximately the same vertical level as the lower end of the inner chuck wall portion **40** where the metal sheet bends to extend radially inwardly and downwardly along the lower portion **42** of the inner chuck wall. The outer chuck wall terminates at an upper edge defined by the edge **34** of the metal sheet that forms the metal end. The upper edge of the outer chuck wall is spaced below the second portion **50** of the skirt **46** in the illustrated embodiment, although it is possible for the upper edge of the outer chuck wall to contact the lower surface of the skirt if desired.

(16) An annular channel **58** is defined between the inner and outer chuck walls. The channel **58** has an annular shape generally matching that of the upper edge **16** of the side wall **14** of the container body. The channel **58** has a radial width selected with regard to the thickness of the side wall at the upper edge **16** such that the side wall can be inserted into the channel with a close fit as shown in FIG. **2**.

(17) A layer of heat-sealable material **60** is disposed on at least the portion of the outer surface of the inner chuck wall **36** that contacts the inner heat-sealable material **26** on the inner surface of the container body side wall **14**. If desired, the heat-sealable layer **60** can cover the entire lower surface of the metal end as shown, which is advantageous for preventing the contents of the container from directly contacting the metal; alternatively, the heat-sealable layer **60** could be localized to the portion of the inner chuck wall's outer surface that contacts the container body side wall, and another coating (e.g., lacquer) could be employed for protecting against metal contact.

(18) At least the portion of the inner surface of the outer chuck wall **56** that contacts the side wall **14** has a layer of heat-sealable material **62** disposed thereon. If desired, the heat-sealable layer **62**

can cover the entire upper surface of the metal end as shown.

(19) Once the metal end **30** is pushed onto the container body side wall **14** to insert the side wall into the channel **58** as shown in FIG. **2**, the outer peripheral region of the metal end **30** is heated by a suitable method (e.g., induction heating) so as to cause the heat-sealable layers **26** and **60** to melt and fuse together and to cause the heat-sealable layers **28** and **62** to melt and fuse together. Upon cooling, the re-solidified heat-sealable material bonds the metal end to the container body side wall in a secure and hermetic fashion. The process of applying the metal end **30** onto the container body and sealing the metal end thereon thus entails no significant deformation of the container body side wall **14** (and particularly no outward rolling or curling of the side wall as required with conventional double-seaming), and no significant deformation of the metal end **30**. The metal end is simply pushed straight onto the side wall, and heating then seals the end onto the container body.

(20) The heat-sealable layers **26**, **28**, **60**, **62** can comprise any of various heat-sealable materials, including but not limited to polyethylene, polypropylene, ionomers such as SURLYN® (ethylene acrylic acid copolymer having acid groups partially neutralized with sodium ions), and the like. The heat-sealable material on the metal end **30** can be, but need not be, identical to the heat-sealable material on the container body side wall **14**, the important consideration being that if different heat-sealable materials are employed, the melting point temperatures of the materials should not be too greatly different.

(21) The present invention is subject to numerous variations and embodiments. In the embodiment of FIGS. **1** and **2**, as noted, the metal end **30** defines a stacking surface **50** on the outer skirt **46**. In accordance with this embodiment, a second container having a bottom metal end **30'** can be stacked atop the top metal end **30** of the first container. In particular, the second container can have a bottom metal end **30'** that is formed generally similarly to the metal end **30** described above. However, unlike the metal end **30**, the bottom metal end **30'** of the second container has its stacking feature defined by the inner chuck wall **36'** rather than by the skirt. More particularly, the metal end **30'** has a bottom rim **44'** formed as an upwardly facing U-shaped section from which the inner chuck wall **36'** extends upwardly. The inner chuck wall has a first portion **37'** just above the bottom rim **44'**, a second portion **39'** above the first portion, and a third portion **40'** above the second portion. The second portion **39'** is closer to being horizontal than the first and third portions and defines a stacking surface for engaging the top rim **44** of the metal end of the underlying container. The bottom rim **44'** has a minimum inside diameter, an outside diameter of the first portion **48** of the outer skirt **46** of the top metal end **30** is less than said minimum inside diameter of the bottom rim **44'**, and an outside diameter of the third portion **52** of the skirt **46** is greater than said minimum inside diameter.

(22) The bottom metal end **30'** also has an outer skirt **46'** that extends upwardly from the bottom rim **44'** to an upper U-shaped section **54'** that faces downwardly, and an outer chuck wall **56'** extends downwardly from the upper U-shaped section and is spaced from the inner chuck wall portion **40'**. An annular channel **58'** is defined between the chuck walls **40'** and **56'** for receiving the side wall **14'** of the second container body, the metal end **30'** being heat-sealed onto the side wall in the manner previously described.

(23) It will be recognized that the embodiment having a top metal end **30** and a bottom metal end **30'** as shown in FIG. **3** can be reversed such that the metal end **30** (or one having a similar configuration in which the stacking feature is defined by the skirt) is used on the bottom end of the container and the metal end **30'** (or one having a similar configuration in which the stacking feature is defined by the inner chuck wall) is used on the top end of the container.

(24) It is also possible to configure the metal end such that there is no stacking feature. Such an embodiment is shown in FIG. **4**. The metal end **130** of this embodiment is generally similar to the metal end **30** previously described, except that the skirt **146** does not have a stacking feature. Additionally, the container body side wall **114** in the embodiment of FIG. **4** is formed of a plastic material rather than a composite material. The plastic material can comprise a thermoplastic

material that is heat-sealable to the heat-sealable layers on the chuck walls of the metal end.

(25) In the embodiments described above, the metal sheet is shaped into its ultimate form prior to being applied to the container body. Thus, the metal end has the skirt, lower U-shaped section, and outer chuck wall already formed in their ultimate configurations before the metal end is pushed onto the container body. Application of the metal end to the container body does not require any specialized tooling for making a double seam, and the difficulties of forming a double seam are avoided.

(26) The embodiments described above and illustrated in the drawings have heat-sealable material layers **26** and **28** on both the inner and outer surfaces of the container body side wall **14**, and correspondingly both chuck walls **36**, **56** have heat-sealable layers **60**, **62** thereon such that two heat seals are formed. However, it is within the scope of the invention to omit one of the heat seals. For example, the heat-sealable layers **26** and **60** can be omitted, such that only one heat seal is formed between the layers **28** and **62** on the outer surface of the container body side wall and the outer chuck wall. Alternatively, the heat-sealable layers **28** and **62** can be omitted, such that only one heat seal is formed between the layers **26** and **60** on the inner surface of the container body side wall and the inner chuck wall.

(27) Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A metal end comprising, prior to application to a container body, a metal sheet having a peripheral edge, the sheet having an outer peripheral region that includes: an inner chuck wall comprising: an upper portion that extends upwardly to a top rim of the metal end, the top rim comprising an outwardly curled, downwardly facing upper U-shaped section, the upper portion being linear when viewed in cross section; a lower portion that is connected to the lower end of the upper portion, the lower portion being linear when viewed in cross section; wherein the upper portion is inclined relative to vertical and is closer to vertical than the lower portion is to vertical; a skirt extending downwardly from the top rim, a lower end portion of the skirt being formed as an inwardly curled, upwardly facing lower U-shaped section, wherein the skirt comprises: a first portion disposed just below the top rim; a second portion disposed below the first portion; and a third portion disposed below the second portion, and wherein the second portion of the skirt is oriented closer to horizontal than the first and third portions; a substantially vertical outer chuck wall having a lower end joined to the lower U-shaped section, the outer chuck wall being linear when viewed in cross section and extending upwardly to the peripheral edge of the metal sheet defining a top edge of the outer chuck wall, wherein the outer chuck wall is spaced from the inner chuck wall so as to define an annular channel therebetween; and heat-sealable material disposed on both of opposing surfaces of the inner and outer chuck walls such that the heat-sealable material allows sealing of the metal end to the container body without significant deformation of the container body.
2. The metal end of claim 1, wherein the second portion forms a stacking surface on which a bottom end of a container can be stacked.
3. The metal end of claim 1, wherein an outside diameter of the first portion is less than an inside diameter of a bottom rim of a bottom metal end of a container.
4. The metal end of claim 1, wherein an outside diameter of the third portion is greater than an

inside diameter of a bottom rim of a bottom metal end of a container.

5. The metal end of claim 1, additionally comprising a substantially horizontal central panel that is connected to a lower end of the inner chuck wall lower portion, the substantially horizontal central panel appearing substantially linear when viewed in cross section.

6. The metal end of claim 1, wherein the lower portion of the inner chuck wall extends radially inwardly and downwardly.

7. The metal end of claim 1, wherein the lower end of the upper portion of the inner chuck wall and the lower end of the outer chuck wall are at approximately the same vertical position.

8. The metal end of claim 7, wherein the approximately the same vertical position is located where the metal sheet bends and extends radially inwardly and downwardly along the lower portion of the inner chuck wall.

9. The metal end of claim 1, wherein the skirt comprises a first portion disposed just below the top rim, a second portion disposed below the first portion, and a third portion disposed below the second portion, and wherein the second portion of the skirt is oriented closer to horizontal than the first and third portions.

10. A metal end comprising, prior to application to a container body, a metal sheet having a peripheral edge, the sheet having an outer peripheral region that includes: an inner chuck wall comprising: an upper portion that extends upwardly to a top rim of the metal end, the top rim comprising an outwardly curled, downwardly facing upper U-shaped section, the upper portion being linear when viewed in cross section; a lower portion that is connected to the lower end of the upper portion, the lower portion being linear when viewed in cross section; wherein the upper portion is inclined relative to vertical and is closer to vertical than the lower portion is to vertical; a skirt extending downwardly from the top rim, a lower end portion of the skirt being formed as an inwardly curled, upwardly facing lower U-shaped section, wherein the skirt comprises: a first portion disposed just below the top rim; a second portion disposed below the first portion; and a third portion disposed below the second portion, and wherein the second portion of the skirt is oriented closer to horizontal than the first and third portions; a substantially vertical outer chuck wall having a lower end joined to the lower U-shaped section, the outer chuck wall being linear when viewed in cross section and extending upwardly to the peripheral edge of the metal sheet defining a top edge of the outer chuck wall, wherein the outer chuck wall is spaced from the inner chuck wall so as to define an annular channel therebetween; and heat-sealable material disposed on at least one of opposing surfaces of the inner and outer chuck walls such that the heat-sealable material allows sealing of the metal end to the container body without significant deformation of the container body, wherein the second portion forms a stacking surface on which a bottom end of a container can be stacked.

11. The metal end of claim 10, wherein an outside diameter of the first portion is less than an inside diameter of a bottom rim of a bottom metal end of a container.

12. The metal end of claim 10, wherein an outside diameter of the third portion is greater than an inside diameter of a bottom rim of a bottom metal end of a container.

13. The metal end of claim 10, additionally comprising a substantially horizontal central panel that is connected to a lower end of the inner chuck wall lower portion, the substantially horizontal central panel appearing substantially linear when viewed in cross section.

14. The metal end of claim 10, wherein the lower portion of the inner chuck wall extends radially inwardly and downwardly.

15. The metal end of claim 10, wherein the lower end of the upper portion of the inner chuck wall and the lower end of the outer chuck wall are at approximately the same vertical position.

16. The metal end of claim 15, wherein the approximately the same vertical position is located where the metal sheet bends and extends radially inwardly and downwardly along the lower portion of the inner chuck wall.

17. The metal end of claim 10, wherein both of the opposing surfaces of the inner and outer chuck walls have the heat-sealable material thereon.
